MSG No. 176.144 $P6_3/m1'$ [Type II, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-6..1'`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,3,5,8,10,12,13,15,17,20,22,24]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[2,4,6,7,9,11,14,16,18,19,21,23]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3..1'`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,3,5,7,9,11,13,15,17,19,21,23]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[2,4,6,8,10,12,14,16,18,20,22,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6..1'`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[1,3,5,8,10,12,13,15,17,20,22,24]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[2,4,6,7,9,11,14,16,18,19,21,23]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6..1'`

No.	position	mapping
1	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[1,3,5,8,10,12,13,15,17,20,22,24]`
2	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[2,4,6,7,9,11,14,16,18,19,21,23]`

Table 5: Wyckoff site: `4e`, site symmetry: `3..1'`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,3,5,13,15,17]`
2	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[2,4,6,14,16,18]`
3	$[0,\, 0,\, -z]$	`[7,9,11,19,21,23]`
4	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	`[8,10,12,20,22,24]`

Table 6: Wyckoff site: `4f`, site symmetry: `3..1'`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	`[1,3,5,13,15,17]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, z+\frac{1}{2}\right]$	`[2,4,6,14,16,18]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$	`[7,9,11,19,21,23]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}-z\right]$	`[8,10,12,20,22,24]`

Table 7: Wyckoff site: `6g`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,7,13,19]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,8,14,20]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,9,15,21]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,10,16,22]`
5	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,11,17,23]`
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,12,18,24]`

Table 8: Wyckoff site: `6h`, site symmetry: `m..1'`

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	`[1,10,13,22]`
2	$\left[x-y, x, \frac{3}{4}\right]$	`[2,11,14,23]`
3	$\left[-y, x-y, \frac{1}{4}\right]$	`[3,12,15,24]`
4	$\left[-x, -y, \frac{3}{4}\right]$	`[4,7,16,19]`
5	$\left[-x+y, -x, \frac{1}{4}\right]$	`[5,8,17,20]`
6	$\left[y, -x+y, \frac{3}{4}\right]$	`[6,9,18,21]`

Table 9: Wyckoff site: `12i`, site symmetry: `11'`

No.	position	mapping
1	$[x, y, z]$	`[1,13]`
2	$\left[x-y, x, z+\frac{1}{2}\right]$	`[2,14]`
3	$[-y, x-y, z]$	`[3,15]`
4	$\left[-x, -y, z+\frac{1}{2}\right]$	`[4,16]`
5	$[-x+y, -x, z]$	`[5,17]`
6	$\left[y, -x+y, z+\frac{1}{2}\right]$	`[6,18]`
7	$[-x, -y, -z]$	`[7,19]`
8	$\left[-x+y, -x, \frac{1}{2}-z\right]$	`[8,20]`
9	$[y, -x+y, -z]$	`[9,21]`
10	$\left[x, y, \frac{1}{2}-z\right]$	`[10,22]`
11	$[x-y, x, -z]$	`[11,23]`
12	$\left[-y, x-y, \frac{1}{2}-z\right]$	`[12,24]`